National University of Lesotho
Department of Mathematics and Computer Science

M3503 Tutorial 2 August 2025

1. Solve the following differential equations.

   (a) $\dfrac{dy}{dx} - y = e^x y^2$

   (b) $\dfrac{dy}{dx} = y(xy^3 - 1)$

   (c) $(-3x + y + 6)dx + (x + y + 2)dy = 0$

   (d) $y' + 2x^2 y - 2xy^2 = 1$; $y_1(x) = x$

   (e) $x\dfrac{dy}{dx} + y = y^2 x^2 \ln x$

   (f) $\dfrac{dy}{dx} + \dfrac{2y}{x} = -x^2 y^2 \cos x$

   (g) $(y + x + 2)dx + (3y - x - 6)dy = 0$

   (h) $(2x - y)dx + (4x + y - 3)dy = 0$

   (i) $y' + y^2 = 1 + x^2; \quad y_1 = x$

   (j) $x\dfrac{dy}{dx} - y(x + 1) = xy^2$

   (k) $x^2\dfrac{dy}{dx} + y^2 = xy$

   (l) $\dfrac{dy}{dx} = x^3(y - x)^2 + \dfrac{y}{x}; \quad y_1 = x$

   (m) $xy' = x^2(\ln x)^2 - 2x^2 \ln x + 1 + 2x^2(1 - \ln x)y + x^2 y^2; \quad y_1 = \ln x$

   (n) $y' + 2y^2 - y = 1$; $y_1(x) = 1$

   (o) $ydx + (2x + 3y - 2)dy = 0$

   (p) $(2y + 4)dx + (x - 1)dy = 0$

2. (a) Show that the substitution $y = y_1 + \dfrac{1}{u}$ reduces the Riccati equation

$$\frac{dy}{dx} = P(x)y^2 + Q(x)y + R(x)$$

   to a linear first-order ordinary differential equation.

   (b) Show that the substitution $y = y_1 + u$ reduces the Riccati equation in 2 (a) above to a Bernoulli equation with $n = 2$.

   (c) Verify that the substitution

$$y = \frac{1}{R(x)z}\frac{dz}{zx}$$

   reduces the Riccati equation in 2 (a) above to the linear homogeneous second-order ordinary differential equation

$$\frac{d^2z}{dx^2} + \left\{P(x) - \frac{R'(x)}{R(x)}\right\}\frac{dz}{dx} - R(x)Q(x)z = 0$$